

Assignment Five: Part C — Reflection Paper

Predictive Modeling for Loan Underwriting Classification

Course: Machine Learning — Fundamentals and Practice

Topic: Predictive Modeling for Loan Underwriting Classification

1. Project Implementation Overview

In this practical assignment, I successfully implemented an end-to-end machine learning classification pipeline designed to automate risk profiling and predict loan approvals using Python and the scikit-learn ecosystem.

The first phase replicated and enhanced the data preprocessing pipeline from Lesson 5 using a raw, unstructured loan dataset (`raw_loan_dataset.csv`). The data required substantial parsing, including scrubbing currency symbols (\$, ,), standardizing noisy user-inputted textual tags (unifying entries like "yse", "y", and "Approved" into clear binary targets), handling missing data points with robust column-level medians and modes, and capping outliers without eliminating rows by using the Interquartile Range (IQR) `.clip()` method.

The major upgrade introduced in this implementation was substituting the standard scaler with `RobustScaler`. Following data cleaning, I engineered non-leaking variables (`DebtToIncome` and `IncomePerYearEmployed`) and built three models: the core baseline Logistic Regression, the ensemble Random Forest Classifier, and an independently researched third classifier — the Support Vector Machine (SVM).

2. Comparison of Models & Sanity Check Evaluation

During the isolated sanity check phase, a single credit-applicant vector was extracted from the holdout testing set (`X_test.iloc[[0]]`) and fed into all three trained algorithms to compare predictions against the ground-truth label.

The models displayed variations in confidence levels and directional outcomes depending on their underlying mathematical design. Logistic Regression and Support Vector Machines align their outputs along structural decision thresholds based on continuous functions — sigmoidal curves and hyperplanes, respectively — while Random Forest generates predictions by polling an assembly of discrete logical trees.

The Random Forest model yielded the most realistic results. Single linear partitions (Logistic Regression) or margin boundaries (SVM) can be overly swayed by minor changes in an individual feature. In contrast, Random Forest relies on a consensus of one hundred diversified decision trees. This ensemble approach handles marginal boundaries much better, preventing a single high-value attribute — such as an unusually high income paired with a low credit score — from forcing an unrealistic prediction.

3. Understanding Random Forest

In my own words, a Random Forest is an ensemble machine learning algorithm that functions by aggregating the individual predictions of many distinct decision trees to produce a single, highly accurate output.

It relies on a technique called Bootstrap Aggregating (Bagging). Instead of training one massive tree on all available data, the algorithm creates dozens or hundreds of trees. Each tree is trained on a random sample of rows and evaluates a random subset of features at each split point. This intentional randomization ensures that the trees grow independently and do not overfit to the same patterns or noise.

When a new loan applicant is evaluated, every tree in the forest processes the data and casts a vote (for example, Tree 1 votes "Approved," Tree 2 votes "Rejected," Tree 3 votes "Approved"). The algorithm counts these votes, and the majority vote determines the final prediction. This collective decision-making dramatically reduces variance and makes the model highly resilient.

4. Alternative Algorithms: Researched Third Classifier (SVM)

For my independent classifier, I chose the Support Vector Machine (SVM).

Why It Fits Loan Prediction

SVMs are built to find an optimal boundary (a hyperplane) that separates two classes with the widest possible safety margin. Loan underwriting is fundamentally a margin-separation problem: banks want to draw a distinct boundary between safe applicants and high-risk defaults. Since the numerical inputs were standardized using a median-centered RobustScaler, the features were well suited to an SVM's distance-based math.

Core Insights Learned

How it Works: SVM maps data vectors into a higher-dimensional space to find a decision hyperplane that maximizes the geometric distance (margin) between the closest data points of the two classes, known as support vectors.

Main Advantage: It is mathematically precise and highly effective in high-dimensional spaces, remaining stable even when the total number of training records is relatively small.

Main Limitation: It does not scale efficiently to massive datasets because calculating the optimal margins becomes computationally expensive. It is also a "black box" model that does not provide easily readable coefficients like Logistic Regression.

Metric Comparison

When evaluating test performance, the SVM model generally outperformed the simple Logistic Regression baseline across all metrics, showing a noticeable reduction in false positives. However, it achieved slightly lower F1-scores than the Random Forest, which was better equipped to handle the categorical features in the dataset.

5. Metrics Discussion & Key Findings

Performance Insights

Random Forest consistently achieved the highest scores across Accuracy, Precision, and the F1-Score. This shows that the ensemble design excels at identifying complex, non-linear relationships, such as how a low credit score might be offset by an exceptionally low Debt-to-Income ratio.

Logistic Regression delivered solid, reliable baseline metrics but fell short in Recall. This suggests that its rigid linear boundary tends to misclassify complex, borderline applicants who fall just outside its strict linear path.

Support Vector Machine (SVM) matched Random Forest closely in Precision, making few false-positive errors. However, its overall Recall was lower, meaning it adopted an overly cautious approach to approving loans.

What This Tells Us About Model Strengths

This performance spread highlights the trade-offs of each architecture. Logistic Regression is fast and transparent but struggles with complex, real-world data patterns. SVM excels at maintaining a clean, conservative boundary (high Precision) but can be overly rigid. Random Forest provides the best all-around performance by blending flexibility with robust error correction, making it highly effective at capturing nuanced applicant profiles.

6. Final Recommendation and Structural Conclusion

For deployment in a real-world financial loan underwriting system, I would select the Random Forest Classifier as the production model. Loan underwriting requires a careful balance between risk mitigation (minimizing defaults) and business growth (maximizing safe loan approvals). The Random Forest model delivers the ideal balance by securing the highest overall F1-Score, demonstrating that it maximizes safe approvals without exposing the institution to undue risk.

Additionally, Random Forest offers a significant operational advantage through its native calculation of Feature Importance. Unlike the abstract mathematical transformations inside an SVM, a Random Forest can explicitly rank which variables — such as CreditScore or DebtToIncome — had the greatest impact on its decisions. This feature allows compliance teams to audit predictions, ensure fair lending practices, and provide transparent explanations to applicants when a loan is denied. This combination of high predictive accuracy and operational transparency makes Random Forest the superior choice for financial deployment.